

1. Update Kali Linux

1. Open a terminal in Kali.[\[geeksforgeeks\]](#)
2. Run:

```
bash
sudo apt update && sudo apt -y full-upgrade
```

This updates packages before installing the web stack.[\[geeksforgeeks\]](#)

2. Install Apache, MariaDB, and PHP

1. In the terminal, install the required services and PHP modules:

```
bash
sudo apt install apache2 mariadb-server php php-mysql php-gd libapache2-mod-php -y
```

Apache provides the web server; MariaDB provides the database; PHP and modules are required for DVWA.[\[geeksforgeeks\]](#)

2. Enable and start services:

```
bash
sudo systemctl enable apache2 mariadb
sudo systemctl start apache2 mariadb
```

This ensures the web server and database start now and on boot.[\[tecnosender\]](#)

3. Download DVWA into the Web Root

1. Change to the Apache web root directory:

```
bash
cd /var/www/html
```

DVWA must reside under the web root to be served by Apache.[\[geeksforgeeks\]](#)

2. Install Git if needed, then clone DVWA:

```
bash
sudo apt install git -y
sudo git clone https://github.com/digininja/DVWA.git
```

This downloads the DVWA source code into `/var/www/html/DVWA`.[\[geeksforgeeks\]](#)

4. Set DVWA Directory Permissions

1. Verify the DVWA directory exists:

```
bash
ls
```

You should see `DVWA` listed.[\[geeksforgeeks\]](#)

2. Set broad permissions so the web server can read and write files for lab use:

```
bash
sudo chmod -R 777 DVWA
```

The video uses `chmod -R 777`; in a real environment you would prefer tighter `www-data` ownership, but for a local test lab this is acceptable.[\[youtube\]](#)[\[geeksforgeeks\]](#)

5. Prepare the DVWA Configuration File

1. Change into the DVWA configuration directory:

```
bash
cd DVWA/config
```

[\[geeksforgeeks\]](#)

2. Copy the default configuration template:

```
bash
sudo cp config.inc.php.dist config.inc.php
```

DVWA uses `config.inc.php` as the active configuration file.[\[geeksforgeeks\]](#)

3. Open `config.inc.php` in a text editor:

```
bash
sudo mousepad config.inc.php
```

(Any editor such as `nano` or `vim` is acceptable; the video uses a GUI editor.)[\[geeksforgeeks\]](#)

4. Set database credentials in the file, for example:

- o Database user: `admin`
- o Database password: `password`

Update the corresponding lines in `config.inc.php`, then save and close.[\[geeksforgeeks\]](#)

6. Create the DVWA Database and User

1. Start a MariaDB shell as root:

```
bash
sudo mysql -u root -p
```

Leave the password blank and press Enter if no root password is configured.[\[youtube\]](#)[\[geeksforgeeks\]](#)

2. Create the DVWA database:

```
sql
CREATE DATABASE dvwa;
```

[\[geeksforgeeks\]](#)

3. Create a dedicated DVWA user and grant privileges, matching the credentials you set in `config.inc.php` (example: `admin / password`):

```
sql
CREATE USER 'admin'@'127.0.0.1' IDENTIFIED BY 'password';
GRANT ALL PRIVILEGES ON dvwa.* TO 'admin'@'127.0.0.1';
FLUSH PRIVILEGES;
EXIT;
```

This allows DVWA to connect to the `dvwa` database from localhost using the specified user.[\[youtube\]](#)[\[geeksforgeeks\]](#)

7. Adjust PHP Configuration (`allow_url_include`)

1. Locate the PHP configuration file under Apache's configuration directory (for example, `/etc/php/8.2/apache2/php.ini`, version may differ).[\[youtube\]](#)[\[geeksforgeeks\]](#)
2. Change to the directory and edit `php.ini`:

```
bash
cd /etc/php/8.2/apache2
sudo mousepad php.ini
```

Replace `8.2` with your installed PHP version if needed.[\[youtube\]](#)[\[digininja-dvwa.mintlify\]](#)

3. Use the editor's search function to find `allow_url_fopen` and `allow_url_include`.[\[youtube\]](#)
4. Ensure both directives are set to `On`:

```
text
allow_url_fopen = On
allow_url_include = On
```

DVWA requires `allow_url_include` for some vulnerabilities to function; the video explicitly enables these flags.[\[geeksforgeeks\]](#)[\[youtube\]](#)

5. Save changes and close the editor.[\[youtube\]](#)

8. Restart Apache

1. Restart Apache to apply the PHP configuration changes:

```
bash
sudo systemctl restart apache2
```

[\[geeksforgeeks\]](#)[\[youtube\]](#)

9. Access DVWA in the Browser

1. Open a web browser inside Kali (for example, Firefox).[\[geeksforgeeks\]](#)
2. Navigate to:

```
text
http://127.0.0.1/DVWA
```

This should load the DVWA login/setup page.[\[youtube\]](#)[\[geeksforgeeks\]](#)

3. If you see a warning about missing database tables, click the **Setup** or **Create / Reset Database** button in the DVWA interface to initialise the schema.[\[digininja-dvwa.mintlify\]](#)

10. Log In and Verify

1. On the DVWA login page, use the default credentials:
 - Username: admin
 - Password: password

[\[digininja-dvwa.mintlify\]](#)

2. After logging in, confirm that DVWA loads the main interface and that modules (such as SQL Injection or XSS) are available.[\[geeksforgeeks\]](#)
3. Optionally set **DVWA Security** to **Low** to begin testing, then raise the level as needed for your experiments.[\[digininja-dvwa.mintlify\]](#)

